Module 1: Unconstrained Extrema

Optimization Techniques

Dr. Muhammed Sabeel K

TKM College of Engineering, Kollam

August 5, 2025

Introduction to Optimization

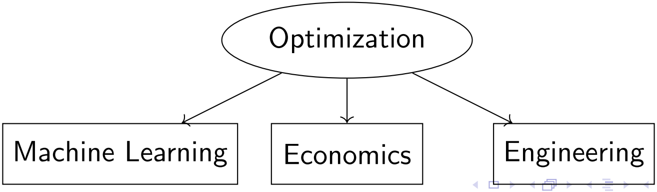
What is Optimization?

Optimization involves finding the best solution from all feasible solutions to maximize or minimize a given objective function.

$$\text{Optimize } f(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n$$

Key Applications

- Machine learning: Minimizing loss functions
- Economics: Maximizing profit or utility
- Engineering: Optimizing design parameters



Module Overview

- Introduction to optimization
- Unconstrained optimization problems
- Maxima and minima (global/local)
- Concavity, inflection, and saddle points
- Necessary and sufficient conditions
- Hessian matrix for extremum classification

Unconstrained Optimization Problems

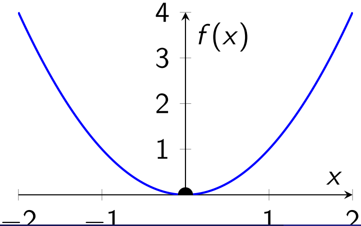
Definition

Finding extreme values (maxima/minima) of a function without any constraints:

$$\text{Optimize } f(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n$$

Example

- Minimizing cost function in machine learning
- Maximizing profit function in economics
- Finding optimal parameters in engineering design



Types of Extrema

Global Extremum

- Highest/lowest value over entire domain
- May not exist for unbounded functions

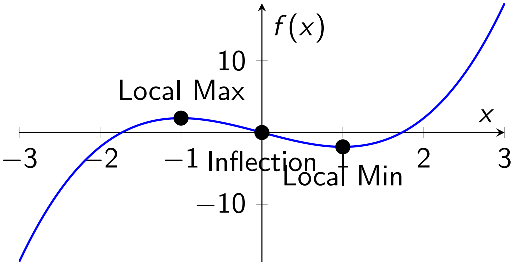
Local Extremum

- Highest/lowest value in neighborhood
- Multiple local extrema may exist

Concavity and Special Points

- Concavity: A function is concave up if its second derivative $f''(x) > 0$, concave down if $f''(x) < 0$.
- Inflection point: Where concavity changes
- Saddle point: Neither max nor min

Visualizing Concavity and Extrema



Necessary Conditions for Extremum

Theorem (First Order Condition)

For a differentiable function $f(x_1, \ldots, x_n)$, a necessary condition for $\mathbf{x}^*$ to be an extremum is:

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

i.e., all partial derivatives vanish at $\mathbf{x}^*$.

Example

For $f(x, y) = x^2 + y^2$:

$$\frac{\partial f}{\partial x} = 2x = 0$$

$$\frac{\partial f}{\partial y} = 2y = 0$$

Stationary point at $(0, 0)$ which is a minimum.

Sufficient Conditions Using Hessian Matrix

Hessian Matrix

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Theorem (Second Order Condition)

At stationary point $\mathbf{x}^*$*:*

- H *positive definite* $\Rightarrow$ *local minimum*
- H *negative definite* $\Rightarrow$ *local maximum*
- H *indefinite* $\Rightarrow$ *saddle point*

Method to Find Inflection Points

Steps

1. Compute the second derivative $f''(x)$.
2. Find points where $f''(x) = 0$ or is undefined.
3. Check the sign of $f''(x)$ on either side of these points to confirm a change in concavity.
4. Verify the point is an inflection point if $f''(x)$ changes sign (e.g., from positive to negative or vice versa).

Note

- Not all points where $f''(x) = 0$ are inflection points; a sign change is required.
- Inflection points can occur at non-critical points ($f'(x) \neq 0$).

Solved Problem: One-Variable Function

Problem

Find and classify all stationary points of:

$$f(x) = x^4 - 4x^2$$

Solution

- Compute the first derivative:

$$f'(x) = 4x^3 - 8x = 4x(x^2 - 2)$$

  Set $f'(x) = 0$:

$$4x(x^2 - 2) = 0 \implies x = 0, x = \sqrt{2}, x = -\sqrt{2}$$

- Compute the second derivative:

$$f''(x) = 12x^2 - 8$$

Classify stationary points:

- At $x = 0$: $f''(0) = -8 < 0 \implies$ local maximum. $f(0) = 0$.
- At $x = \sqrt{2}$: $f''(\sqrt{2}) = 12(\sqrt{2})^2 - 8 = 24 - 8 = 16 > 0 \implies$ local minimum. $f(\sqrt{2}) = (\sqrt{2})^4 - 4(\sqrt{2})^2 = 4 - 8 = -4$.
- At $x = -\sqrt{2}$: $f''(-\sqrt{2}) = 16 > 0 \implies$ local minimum. $f(-\sqrt{2}) = -4$.

Solved Example: Finding an Inflection Point

Problem

Find the inflection points of the function:

$$f(x) = x^3 + x$$

Solution

1. Compute the first derivative:

$$f'(x) = 3x^2 + 1$$

   Note: $f'(x) \geq 1 > 0$, so there are no critical points ($f'(x) \neq 0$).

2. Compute the second derivative:

$$f''(x) = \frac{d}{dx}(3x^2 + 1) = 6x$$

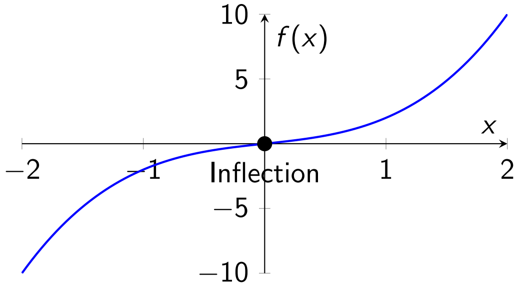
3. Set $f''(x) = 0$:

$$6x = 0 \implies x = 0$$

Visualization of the Example

Graph of $f(x) = x^3 + x$

The inflection point at $x = 0$ marks the transition from concave down to concave up.



Solved Problem 1: Classifying Extrema

Problem

Find and classify all stationary points of:

$$f(x, y) = x^3 - 3xy^2 + y^3$$

Solution

1. Find partial derivatives:

$$f_x = 3x^2 - 3y^2$$

$$f_y = -6xy + 3y^2$$

2. Set $\nabla f = 0$:

$$3x^2 - 3y^2 = 0 \Rightarrow x = \pm y$$

$$-6xy + 3y^2 = 0 \Rightarrow y(-2x + y) = 0$$

Solved Problem 1 (Continued)

Three cases:

1. $y = 0 \Rightarrow x = 0$: Point $(0, 0)$
2. $x = y \Rightarrow -6y^2 + 3y^2 = 0 \Rightarrow y = 0$ (same as above) or $y = 2$: Point $(2, 2)$
3. $x = -y \Rightarrow 6y^2 + 3y^2 = 0 \Rightarrow y = 0$ (same as above)

Only stationary points: $(0, 0)$ and $(2, 2)$

Solved Problem 1 (Classification)

Compute second derivatives:

$$f_{xx} = 6x, \quad f_{xy} = -6y, \quad f_{yy} = -6x + 6y$$

Hessian matrix:

$$H = \begin{bmatrix} 6x & -6y \\ -6y & -6x + 6y \end{bmatrix}$$

At $(0, 0)$:

$$H = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Test fails - need higher order analysis (actually a saddle point)
At $(2, 2)$:

$$H = \begin{bmatrix} 12 & -12 \\ -12 & 0 \end{bmatrix}$$

Determinant $= -144 < 0 \Rightarrow$ Saddle point

Solved Problem 2: Quadratic Function

Problem

Find and classify all stationary points of:

$$f(x, y) = 2x^2 + y^2 - 4xy$$

Solution

1. Find partial derivatives:

$$f_x = 4x - 4y$$

$$f_y = 2y - 4x$$

2. Set $\nabla f = 0$:

$$4x - 4y = 0 \Rightarrow x = y$$

$$2y - 4x = 0 \Rightarrow y = 2x$$

Combine: $x = 2x \Rightarrow x = 0, y = 0$
Stationary point: $(0, 0)$

Solved Problem 2 (Classification)

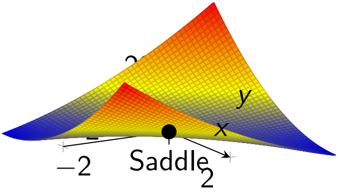
Compute second derivatives:

$$f_{xx} = 4, \quad f_{xy} = -4, \quad f_{yy} = 2$$

Hessian matrix:

$$H = \begin{bmatrix} 4 & -4 \\ -4 & 2 \end{bmatrix}$$

At $(0, 0)$: Determinant $= 4 \cdot 2 - (-4)^2 = 8 - 16 = -8 < 0 \Rightarrow$ Saddle point



Solved Problem: Two-Variable Function

Problem

Find and classify all stationary points of:

$$f(x, y) = x^2 + y^2 + xy - x - y$$

Solution

1. Compute partial derivatives:

$$f_x = 2x + y - 1$$

$$f_y = 2y + x - 1$$

Set $\nabla f = 0$:

$$2x + y = 1$$

$$x + 2y = 1$$

Solve: Subtract second from first: $(2x + y) - (x + 2y) = 1 - 1 \implies x - y = 0 \implies x = y$. Substitute $x = y$ into $2x + y = 1$: $2x + x = 3x = 1 \implies x = \frac{1}{3}, y = \frac{1}{3}$. Stationary point: $\left(\frac{1}{3}, \frac{1}{3}\right)$.

2. Compute second derivatives:

$$f_{xx} = 2, \quad f_{xy} = 1, \quad f_{yy} = 2$$

Hessian matrix:

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

3. Classify: Determinant $\det(H) = 2 \cdot 2 - 1 \cdot 1 = 4 - 1 = 3 > 0$, and $f_{xx} = 2 > 0$. Thus, H is positive definite $\implies$ local minimum at $\left(\frac{1}{3}, \frac{1}{3}\right)$. Value:

$$f\left(\frac{1}{3}, \frac{1}{3}\right) = \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \frac{1}{3} \cdot \frac{1}{3} - \frac{1}{3} - \frac{1}{3} = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} - \frac{1}{3} - \frac{1}{3} = \frac{3-6}{9} = -\frac{1}{3}$$

Solved Problem: Three-Variable Function

Problem

Find and classify all stationary points of:

$$f(x, y, z) = x^2 + y^2 + z^2 + xy + yz$$

Solution

1. Compute partial derivatives:

$$f_x = 2x + y$$

$$f_y = 2y + x + z$$

$$f_z = 2z + y$$

   Set $\nabla f = 0$:

$$2x + y = 0$$

$$x + 2y + z = 0$$

$$y + 2z = 0$$

   From third: $y = -2z$. Substitute into first: $2x - 2z = 0 \implies x = z$. Substitute $x = z$, $y = -2z$ into second: $z + 2(-2z) + z = -2z = 0 \implies z = 0$. Thus, $x = 0$, $y = 0$. Stationary point: $(0, 0, 0)$.

2. Compute second derivatives:

$$f_{xx} = 2, \quad f_{xy} = 1, \quad f_{xz} = 0, \quad f_{yy} = 2, \quad f_{yz} = 1, \quad f_{zz} = 2$$

   Hessian matrix:

Application Problem 1: Cost Minimization (Engineering)

Problem

A company designs a cylindrical container with volume 1000 cm^3. The cost of material is \$0.02/cm^2 for the top and bottom, and \$0.01/cm^2 for the side. Find the radius r that minimizes the total cost.

Formulation

- Volume: $\pi r^2 h = 1000 \implies h = \frac{1000}{\pi r^2}$.
- Surface area: Top/bottom $= 2\pi r^2$, side $= 2\pi rh$.
- Cost function:

$$C(r) = 0.02 \cdot 2\pi r^2 + 0.01 \cdot 2\pi r \cdot \frac{1000}{\pi r^2} = 0.04\pi r^2 + \frac{20}{r}$$

- Domain: $r > 0$.

Problem 1: Solution

Solution

1. Compute the first derivative:

$$C'(r) = 0.08\pi r - \frac{20}{r^2}$$

Set $C'(r) = 0$:

$$0.08\pi r = \frac{20}{r^2} \implies 0.08\pi r^3 = 20 \implies r^3 = \frac{20}{0.08\pi} \approx 79.577$$

$$r \approx (79.577)^{1/3} \approx 4.3\,\text{cm}$$

2. Compute the second derivative:

$$C''(r) = 0.08\pi + \frac{40}{r^3}$$

At $r \approx 4.3$: $r^3 \approx 79.577$, so $\frac{40}{r^3} \approx \frac{40}{79.577} \approx 0.502$, and $0.08\pi \approx 0.251$.

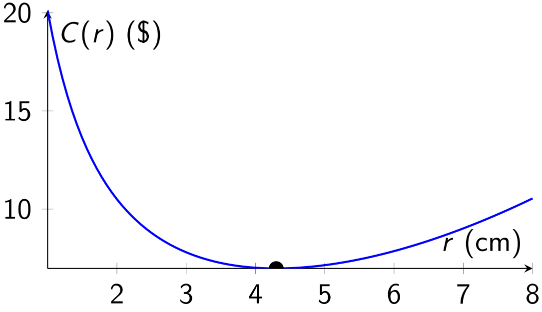
$$C''(4.3) \approx 0.251 + 0.502 = 0.753 > 0 \implies \text{local minimum}$$

3. Verify cost: $C(4.3) \approx 0.04\pi(4.3)^2 + \frac{20}{4.3} \approx 2.31 + 4.65 = 6.96\,\$$.

Problem 1: Visualization

Graph of $C(r) = 0.04\pi r^2 + \frac{20}{r}$

Minimum cost at $r \approx 4.3\,\text{cm}$.



Application Problem 2: Profit Maximization (Economics)

Problem

A company produces two products, A and B, with profit function:

$$P(x, y) = 100x - x^2 + 80y - y^2 - xy$$

where x and y are the units produced of A and B (in thousands). Find the production levels that maximize profit.

Formulation

- Objective: Maximize $P(x, y) = 100x - x^2 + 80y - y^2 - xy$.
- Domain: $x \geq 0, y \geq 0$, but assume unconstrained for simplicity (optimal point lies in first quadrant).

Problem 2: Solution

Solution

1. Compute partial derivatives:

$$P_x = 100 - 2x - y$$

$$P_y = 80 - 2y - x$$

Set $\nabla P = 0$:

$$100 - 2x - y = 0 \implies 2x + y = 100$$

$$80 - 2y - x = 0 \implies x + 2y = 80$$

Solve: Multiply second by 2: $2x + 4y = 160$. Subtract first: $(2x + 4y) - (2x + y) = 160 - 100 \implies 3y = 60 \implies y = 20$. Substitute $y = 20$ into $x + 2y = 80$: $x + 40 = 80 \implies x = 40$. Stationary point: $(40, 20)$.

2. Compute second derivatives:

$$P_{xx} = -2, \quad P_{xy} = -1, \quad P_{yy} = -2$$

Hessian:

$$H = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}$$

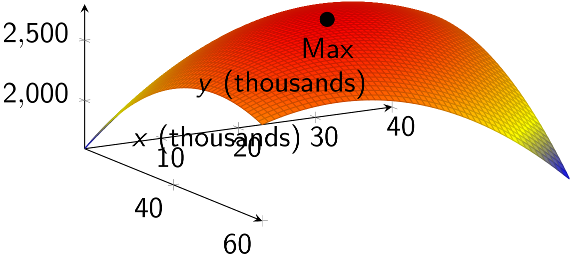
Determinant: $\det(H) = (-2)(-2) - (-1)(-1) = 4 - 1 = 3 > 0$, and $P_{xx} = -2 < 0$. H is negative definite $\implies$ local maximum.

3. Profit: $P(40, 20) = 100(40) - 40^2 + 80(20) - 20^2 - 40 \cdot 20 = 4000 - 1600 + 1600 - 400 - 800 = 2800$.

Problem 2: Visualization

Graph of $P(x, y) = 100x - x^2 + 80y - y^2 - xy$

Maximum profit at $(40, 20)$.



Application Problem 3: Loss Minimization (Machine Learning)

Problem

In a linear regression model $y = w_0 + w_1 x$, minimize the mean squared error (MSE) for data points: (1, 2), (2, 4), (3, 5). The loss function is:

$$L(w_0, w_1) = \frac{1}{6} \sum_{i=1}^{3} (w_0 + w_1 x_i - y_i)^2$$

Formulation

- Objective: Minimize $L(w_0, w_1) = \frac{1}{6}[(w_0 + w_1 \cdot 1 - 2)^2 + (w_0 + w_1 \cdot 2 - 4)^2 + (w_0 + w_1 \cdot 3 - 5)^2]$.
- Domain: $w_0, w_1 \in \mathbb{R}$.

Problem 3: Solution

Solution

1. Compute partial derivatives:

$$L = \frac{1}{6}[(w_0 + w_1 - 2)^2 + (w_0 + 2w_1 - 4)^2 + (w_0 + 3w_1 - 5)^2]$$

$$\frac{\partial L}{\partial w_0} = \frac{1}{6} \cdot 2[(w_0 + w_1 - 2) + (w_0 + 2w_1 - 4) + (w_0 + 3w_1 - 5)] = \frac{1}{3}(3w_0 + 6w_1 - 11)$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{6} \cdot 2[(w_0 + w_1 - 2) \cdot 1 + (w_0 + 2w_1 - 4) \cdot 2 + (w_0 + 3w_1 - 5) \cdot 3]$$

Simplify: $\frac{\partial L}{\partial w_1} = \frac{1}{3}(6w_0 + 14w_1 - 25)$. Set $\nabla L = 0$:

$$3w_0 + 6w_1 = 11$$

$$6w_0 + 14w_1 = 25$$

Solve: Multiply first by 2: $6w_0 + 12w_1 = 22$. Subtract: $(6w_0 + 14w_1) - (6w_0 + 12w_1) = 25 - 22 \implies 2w_1 = 3 \implies w_1 = 1.5$. Substitute $w_1 = 1.5$ into $3w_0 + 6 \cdot 1.5 = 11$: $3w_0 + 9 = 11 \implies 3w_0 = 2 \implies w_0 = \frac{2}{3}$. Stationary point: $\left(\frac{2}{3}, 1.5\right)$.

2. Hessian:

$$L_{w_0 w_0} = \frac{1}{3} \cdot 3 = 1, \quad L_{w_0 w_1} = \frac{1}{3} \cdot 6 = 2, \quad L_{w_1 w_1} = \frac{1}{3} \cdot 14 = \frac{14}{3}$$

$$H = \begin{bmatrix} 1 & 2 \\ 2 & \frac{14}{3} \end{bmatrix}$$

Key Observations

- At $x = 0$, $f'(0) = 1 \neq 0$, so the inflection point is not a critical point.
- The function $f(x) = x^3 + x$ is always increasing ($f'(x) > 0$), but its concavity changes at $x = 0$.
- This example shows that inflection points can occur independently of maxima or minima.

Exercise Questions

Individual Exercise 1

For the function $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$:

1. Find all stationary points
2. Classify each point using the Hessian matrix
3. Identify any global minima/maxima

Individual Exercise 2

For the function $f(x, y) = x^2y + y^3 - 3x^2 - 3y^2$:

1. Find all stationary points
2. Classify each point using the Hessian matrix
3. Sketch a diagram indicating the nature of each point

Team Exercises

Team Exercise 1

Your team is given:

$$f(x,y) = e^{-(x^2+y^2)} + 2e^{-((x-1)^2+(y-1)^2)}$$

1. Find all stationary points numerically
2. Classify each point
3. Visualize the function and mark the extrema
4. Discuss real-world applications where such functions might arise

Team Exercise 2

Consider the function:

$$f(x,y) = x^3 + y^3 - 3xy$$

Summary

- Optimization seeks the best solution for an objective function
- Unconstrained optimization focuses on extrema without constraints
- First derivatives identify stationary points
- Hessian matrix classifies the nature of these points
- Applications in machine learning, economics, and engineering

Next Steps

- Practice identifying and classifying stationary points
- Explore numerical methods for finding extrema
- Prepare for constrained optimization (next module)

References

- Taha, H. A., Operations Research: An Introduction (10th ed.)
- Rao, S. S., Engineering Optimization: Theory and Practice
- NPTEL courses on Optimization Techniques